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Lanolin-derived branched-chain fatty acids supplementation in early-life Holstein calves

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ABSTRACT

This study was conducted to explore the effects of branched-chain fatty acids derived from lanolin (BCFA-DFL) on growth performance, digestibility, blood metabolites, intestinal development, and fecal bacterial structure of calves aged 0 to approximately 28 d. Twenty-four newborn Holstein dairy calves (birth weight = 41.88 ± 4.56 kg, mean ± SD) were divided into the control group receiving whole milk without BCFA-DFL and the treatment group (BCFA) receiving whole milk with BCFA-DFL (4 g/d) supplementation according to the randomized block design based on their birth weight and birth date. Calves fed BCFA-DFL had higher ADG, feed efficiency, heart girth, and total-tract apparent digestibility of nutrients, but the treatments did not affect total DMI. Plasma cholesterol and haptoglobin were lower in calves fed BCFA-DFL than in the control group. The BCFA-DFL supplementation reduced the frequency of diarrhea, the duration of diarrhea, and the number of medication doses required for treating diarrhea. In contrast to the control group, feeding BCFA-DFL increased the gene relative expression of tight junction protein 1, claudin 2, β-defensin-1, β-defensin 4, regenerating islet-derived protein (REG) 3α, REG 3γ, and IL-10 in the small intestine of calves, while decreasing the gene relative expression of IL-1β and TNF-α. Supplementation with BCFA significantly elevated the total content of short-chain fatty acids (SCFA) in calf feces, as well as the proportions of isobutyric acid and butyric acid relative to total SCFA. In addition, the relative abundance of *Subdoligranulum* and *Faecalibacterium* was significantly increased in the feces of calves fed BCFA-DFL at 14 d of age, and *Prevotella_9* was the first to appear in the BCFA group at 14 d of age. Overall, this study explored the effect of BCFA-DFL on the health status of calves in the early stage of their lives, laying a foundation for the research on BCFA-DFL in terms of the intestinal health of young ruminants.

Key words: branched-chain fatty acids, calf performance, intestinal digestion and metabolism, intestinal health

INTRODUCTION

Diarrhea is the primary cause of morbidity in preweaning calves, particularly during the first month of life, and constitutes a major factor threatening their survival rate (Mõtus et al., 2018; Pempek et al., 2019). Calves at approximately 2 wk of age undergo a critical transition from passive to active immunity, during which their immune status is most compromised, rendering them highly susceptible to diarrhea (Hulbert and Moisá, 2016; Shivley et al., 2018; Urie et al., 2018). During this period, the intestinal microbiota of calves begins to colonize and interact with mucosal immunity (Dias et al., 2018; Fan et al., 2021; Du et al., 2023). The occurrence of diarrhea can lead to structural and functional disorders of the microbiota (Antharam et al., 2013), impair calf growth, negatively affect adult conception rates and productivity (Aghakeshmiri et al., 2017; Moreira et al., 2026), and ultimately constrain the economic development of livestock operations. Therefore, implementing effective measures to improve intestinal health and reduce diarrhea incidence in early-life calves is of utmost urgency.

The establishment of a healthy intestinal microbiota in early life is closely associated with the development of intestinal immunity. Notably, host immune homeostasis has been identified as a core driver of intestinal microbial colonization in preweaning calves (Nishihara et al., 2024; Huang et al., 2025). Therefore, nutritional interventions targeting the early gut microbiota and mucosal immunity have emerged as a promising strategy for the prevention and treatment of early-life calf intestinal disorders. Branched-chain fatty acids (**BCFA**) are a class of fatty acids with a methyl group attached to the second (iso-) or third (anteiso-) carbon at the methyl end of the carbon chain (Kaneda, 1977). They possess anti-inflammatory properties, are widely distributed in trace amounts in the tissues and body fluids of humans

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and ruminants, and serve as components of bacterial lipids (Ran-Ressler et al., 2014; Dingess et al., 2017; Gozdzik et al., 2023). Previous studies have confirmed in human intestinal epithelial cells (in vitro) and mouse models that BCFA prevent and alleviate intestinal inflammation by regulating the expression of inflammatory cytokines, for example IL-1β, IL-8, and IL-10 (Yan et al., 2017, 2018; Zhang et al., 2024). Additionally, Ran-Ressler et al. (2011) demonstrated that BCFA not only upregulated intestinal IL-10 expression in mice but also promoted the intestinal colonization of *Bacillus subtilis*. Coincidentally, 90% of the lipids in *Bacillus subtilis* are BCFA, and it has long been recognized as a symbiotic bacterium of BCFA (Kaneda, 1977). Our team previously found that fecal BCFA, including anteiso-C15:0, iso-C16:0, iso-C17:0, iso-C18:0, and even-BCFA (iso-C14:0, iso-C16:0, and iso-C18:0), can serve as marker factors to distinguish healthy calves from diarrheic ones, and their levels are positively correlated with the relative abundance of *Subdoligranulum* (Xin et al., 2021). *Subdoligranulum* is an intestinal probiotic but has not yet been successfully cultured in vitro. Therefore, it remains unclear whether exogenous BCFA supplementation can affect its intestinal colonization (Nan et al., 2024). The rumen of newborn calves is underdeveloped, and their nutrient absorption mode is analogous to that of monogastric animals (Kumar et al., 2021). However, due to interspecies differences, the regulatory effect of BCFA on calf intestinal health and growth performance remains to be explored through direct in vivo studies in calves. Meanwhile, our team's latest research demonstrated that BCFA can mitigate the LPS-induced upregulation of inflammatory factor expression (e.g., IL-1β, IL-8, TNF-α, and INF-γ) in calf small intestinal epithelial cells (Zhang et al., 2025). This study represents a rare exploration of BCFA applied directly to calf intestinal cells, which has enhanced our confidence in translating this intervention to in vivo applications in calves.

However, it is difficult to enrich highly pure BCFA, which leads to an extremely high price. Therefore, as we have noted, only a few studies on BCFA have been conducted at the cellular level or in mice. Lanolin is a byproduct of the wool washing industry and is relatively cheap and rich in BCFA (Yao and Hammond, 2006; Wang et al., 2020). In recent years, the food and chemical industry have been committed to the separation and enrichment of high-purity BCFA from lanolin for application, and a relatively complete process has been established (Yao and Hammond, 2006; Wang et al., 2020). Therefore, we extracted and enriched BCFA derived from lanolin (**BCFA-DFL**) and fed it to calves, aiming to explore the effects of BCFA-DFL on the diarrhea rate and intestinal health of calves and conduct a comprehensive evaluation of the growth performance, digestion, and metabolism of calves.

MATERIALS AND METHODS

Extraction and Analysis of BCFA-DFL

All BCFA used in this study were prepared in the Ruminant Nutrition Laboratory of Northeast Agricultural University (Harbin, China). The main preparation method was as described by Wang et al. (2020). Briefly, lanolin undergoes sequential saponification and calcification reactions to produce calcium soap of fatty acids. The calcium soap is then mixed with anhydrous ethanol and heated multiple times to eliminate unsaponified products. The purified calcium soap is acidified with hydrochloric acid, followed by sequential extraction and distillation using n-hexane to isolate lanolin fatty acid (**LFA**). The LFA were combined with urea and 95% ethanol in a 1:2:6 ratio (mass/volume/mass) and heated at 60°C until the solution turned pale yellow. The mixture was then sealed and incubated at 4°C for 12 h to ensure complete reaction of the fatty acid with urea, resulting in a complex. Following full crystallization and separation, the liquid phase was extracted thoroughly with petroleum ether to yield BCFA-DFL. Methyl esterification of fatty acids was performed according to the methods described by Raes et al. (2001) and Xin et al. (2021). The composition of BCFA-DFL was analyzed using a Shimadzu Nexis GC-MS system (GC2030/QP2020NX, Japan). All derivative samples were injected into the GC-MS system and separated using a DB-5 capillary column (30 m × 0.25 mm × 0.25 μm, Agilent). Chromatograph acquisition and MS identification were performed by chromatography-MS software (LabSolutions, Shimadzu, Japan). The chemical structure of each fatty acid was confirmed by searching the National Institute of Standards and Technology spectrum library through the chemical workstation data processing system for spectrum analysis (Massart-Leën et al., 1981; Hamerly et al., 2015). Finally, the area normalization method was used for the quantification of fatty acids.

Preparation of the BCFA-DFL Nanoemulsion

The hydrophobic nature of the BCFA-DFL extracted in this study hinders its use in calf diets. Nanoemulsion systems have been recognized as effective carriers for hydrophobic compounds in aqueous settings (Shi et al., 2022). Consequently, the fatty acids were transformed into nanoemulsions to facilitate mixing in whole milk for feeding. The emulsifier was formulated with Tween-80 and soy lecithin 3:2 (mass/mass). The BCFA-DFL, sterile water, and emulsifier were evenly mixed at a ratio of 10:89:1 (mass/vol/mass) and then made into a uniform emulsion using a homogenizer (FA30D, FLUKO, Shanghai, China) at 15,000 rpm/min (Mehmood et al., 2019).

Furthermore, the water and emulsifier were prepared in a ratio of 99:1 (mass/vol) using the same method to create the emulsifier solution.

Animals, Experimental Design, and Diets

This study was conducted in Zhongye Pasture (46.93°N, 124.81°E; Daqing, China) from June to July 2024. All procedures involving animals in the experiment received approval from the Animal Ethics Committee of Northeast Agricultural University. The project application code is No. NEAUEC20240293.

Twelve female calves and 12 male calves were transferred to individual pens (1.22 m × 1.8 m) bedded with straw immediately after birth. Calves were transferred to an outdoor calf hutch bedded with sand and rice husks for rearing 24 h after birth. Newborn calves were fed colostrum according to the colostrum management protocol of the ranch: 4 L of colostrum was administered within 2 h of birth, and another 2 L within 6 h of birth. All feedings were administered via an esophageal tube. All colostrum was analyzed before feeding using a portable refractometer (Nasco, Kellogg, ID). If the Brix value was higher than 22%, it indicated that the quality of colostrum met the feeding requirements; otherwise, it was discarded. Serum IgG levels (≥10 g/L) were measured 24 h after colostrum feeding as an indicator of health to assess the passive transfer of immunity. Twenty-four calves were used in a completely randomized block design and randomly assigned to 2 treatment groups of 6 males and 6 females according to birth weight and birth date that were fed whole milk without BCFA-DFL (control, n = 12; 41.58 ± 4.12 kg birth weight, mean ± SD) or 4 g BCFA-DFL in the whole milk daily (BCFA, n = 12; 42.17 ± 4.60 kg birth weight, mean ± SD). Research on BCFA-DFL in ruminants has not yet been reported. Given that the digestive characteristics of newborn calves resemble those of monogastric animals, reference dosages in studies exploring the application of certain components in calves are often based on monogastric models (Maciej et al., 2016). In addition, it has become common practice to use mice as model organisms in conjunction with calves in such research (He et al., 2022; Zhong et al., 2026). Accordingly, the amounts of BCFA-DFL fed to the calves in this study were based on our previous study where it was applied to C57BL/6J mice (J. Y. Lv, Y. Cao, Y. B. Zhu, and H. S. Xin, Northeast Agricultural University, Harbin, China, unpublished). Our results showed that supplementation of BCFA-DFL at 100 mg/kg of BW enhanced immunoglobulin levels in the blood and effectively suppressed the upregulation of intestinal inflammation-related gene (e.g., IL-1β and IL-8) expression induced by *Escherichia coli* K99. Therefore, BCFA-DFL was added at a fixed amount of 4 g/d according to the average BW of the calves at the time of inclusion in the experiment. The BCFA-DFL was fed daily in the form of an emulsion (40 mL, 0.1 g/mL) mixed into whole milk during the morning feeding. The control group was fed with the same amount of emulsifier solution. During the experimental period, calves were fed whole pasteurized milk at 0600 and 1800 h each day. The amount of feeding was controlled through the Calf Feeding Management system (Yinchuan, China) according to the feeding procedure (Figure 1). Refusals were recorded after each meal. Calves in the trial were fed starter from 3 d of age. The starter was provided by Yuanxing Co. Ltd. (Inner Mongolia Autonomous Region, China). Calf starter and water were provided ad libitum during the whole experimental period.

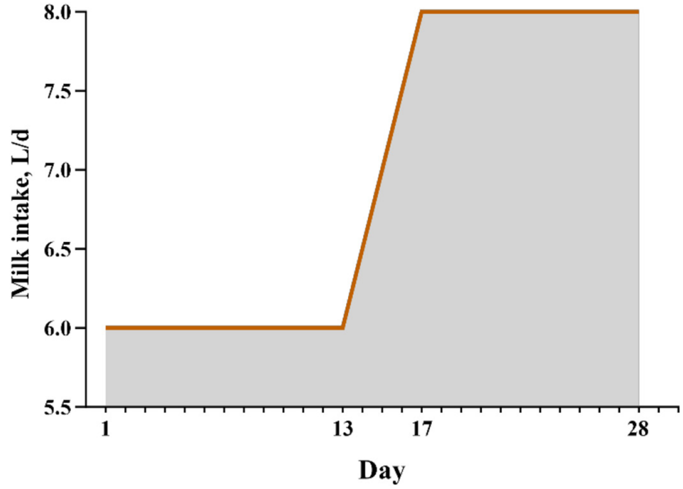

Figure 1. The whole milk feeding program for claves of varying ages.

Sample and Data Collection

Diet Collection. Starter pellets were sampled for analysis on d 3 of the experiment. To obtain a representative sample of fresh starter, 3 samples were collected from 3 different bags. Whole milk was collected at 4-d intervals during the experimental period (specifically on d 4, 8, 12, 16, 20, 24, and 28). The refusal of starter was recorded daily before the morning feeding, and the refusal of whole milk was recorded twice a day. Feed efficiency was determined as the ratio of BW gain to total DMI (milk solids and starter intake).

Assessment of Health Status. The health of the calves was checked daily during the experimental period by the farm veterinarian and a member of the research team before morning feeding. At the same time, cough, respiratory rate, feces with blood, feces score, and rectal temperature were recorded for each calf. Fecal scores were monitored daily according to a 4-point fecal consistency scoring system: score 1 = firm; score 2 = soft or moderate consistency; score 3 = runny or mild diarrhea; and score 4 = watery or profuse diarrhea (Xu et al., 2022). Calves with fecal scores higher than 2 were referred to veterinary care according to the procedures at the farm. The hydration therapy method was employed in the initial stages of calf diarrhea. For diarrhea that persisted for 2 d or longer, treatment included the administration of Chinese herbal preparations and antibiotics. Diarrhea frequency was calculated with the following equation: diarrhea frequency = [(number of diarrhea calves × days of diarrhea)/(total number of calves × days of trial)] × 100%.

BW and Frame Size Measurement. Body weight, withers height (the distance from the ground to a point directly above the most-caudal point of the scapula), body length (the distance from the sternum to the aitchbone), heart girth (chest circumference measured just behind the elbows), and hip height (the distance from the ground to the top of the hip bones) were measured at 0 (immediately after birth), 14, and 28 d of age. All measurements at 14 and 28 d of age were conducted before the morning feeding. The ADG was calculated as weight gain divided by the number of days.

Blood and Fecal Collection. Jugular vein whole blood was collected from calves before morning feeding at 14 and 28 d of age in 3 vacuum-collection tubes containing K_2EDTA, then all tubes were centrifuged at 3,000 × g for 15 min at 4°C. All plasma samples were stored at −20°C until analysis.

Fecal samples were obtained from all calves before the morning feeding at 14 and 28 d of age. Approximately 5 g of each sample were promptly frozen in liquid nitrogen, then preserved at −80°C until bacterial communities were determined. Another 5 g of feces were added to 0.1 mL of 50% sulfuric acid and stored at −20°C until short-chain fatty acids (**SCFA**) were determined (Kazemi-Bonchenari et al., 2017). Before morning feeding at the age of 26 d, the feces of all calves were collected at 6-h intervals (0600, 1200, 1800, and 2400 h) for 3 d to determine apparent digestibility (Xu et al., 2022). Strict hygiene procedures were performed to minimize contamination during sampling.

Tissue Sampling. In ruminant production, female calves are typically retained for replacement heifers. Moreover, many published studies preferentially select male calves for slaughter to collect tissue samples (Jarrige et al., 2020; Liu et al., 2022; Nicola et al., 2023). Therefore, all male calves were slaughtered at 29 d of age. The entire gastrointestinal tract was removed and placed on the operating table, and the duodenum specimen was taken 10 cm distal to the pyloric sphincter. The jejunum was defined as 100 cm distal to the pyloric sphincter, and a 10-cm segment of intestine was collected. Ileum samples were collected 30 cm proximal to the ileocecal junction (Liang et al., 2016; Van Soest et al., 2022). Tissue samples were cut longitudinally and washed with sterile PBS. The cross-sections were fixed in a 4% formaldehyde solution, and another cross-section was flash-frozen in liquid nitrogen, after which samples were stored at −80°C for analysis.

Table 1. Nutrient composition of whole milk and starter to calves

Item	Whole milk	Starter
Ingredient, % of DM		
Wheat bran		5.92
Steam-flaked corn		31.59
Cane molasses		1.53
Soybean meal		14.63
Extruded soybean		5.05
Canola meal		4.78
Corn gluten		4.46
Wheat shorts		5.12
Soybean hull		21.49
Palm fat		1.50
Calves starter premix[1]		3.93
Total		100.00
Nutrient composition,[2] %		
DM	12.29	88.69
CP	3.25	19.73
EE	4.40	5.19
Lactose	4.63	—
NDF		27.98
ADF		12.80
Ash		8.39

[1]The premix provided per kilogram of the starter was as follows: vitamin A, 13,050 IU; vitamin D, 3,262 IU; vitamin E, 260,997 IU; Fe, 116,817 mg; Cu, 19,621 mg; Mn 48,516 mg; Zn 74,603 mg; Se, 0.746 mg; I, 1,343 mg; Co, 0.966 mg.

[2]The nutritive values are the means of the results of the analysis of samples collected each time.

Sampling Analyses

Feed and Feces. All whole milk samples collected were analyzed for milk composition (milk fat, milk protein, lactose, and total solids) using a milk composition analyzer (FOSS, Denmark). Individual fecal samples were synthesized into 1 sample per calf. Starter samples and fecal samples were dried at 55°C for 72 h. Starter and feces were crushed in a grinder (Dingli Machinery Equipment Co. Ltd., Zhejiang, China) through a 1-mm screen. The apparent digestibility of nutrients in the whole tract was evaluated using acid-insoluble ash as an internal marker (Liu et al., 2020a). The contents of DM (method 930.15), ADF (method 973.18), CP (method 990.03), ether extract (**EE**; method 2003.05), and ash (method 942.05) were measured according to the standard procedures of AOAC International (2000). Heat-stable α-amylase and sodium sulfite were used for the NDF procedure according to Van Soest et al. (1991). The chemical composition of whole milk and starter is shown in Table 1.

Blood Analyses. Plasma total protein, albumin, triglycerides, cholesterol, low-density lipoprotein cholesterol

Table 2. Composition and content (mean ± SD) of branched-chain fatty acids derived from lanolin (BCFA-DFL)

Fatty acid	Content (% of total fatty acids)
iso-C10:0	0.77 ± 0.05
C10:0	0.83 ± 0.09
anteiso-C11:0	2.66 ± 0.11
C11:0	0.17 ± 0.02
iso-C12:0	3.83 ± 0.17
C12:0	1.02 ± 0.01
iso-C13:0	0.45 ± 0.01
anteiso-C13:0	4.94 ± 0.16
C13:0	0.57 ± 0.05
iso-C14:0	14.47 ± 0.72
C14:0	5.49 ± 0.68
iso-C15:0	2.71 ± 0.02
anteiso-C15:0	16.56 ± 0.12
C15:0	1.52 ± 0.20
iso-C16:0	14.27 ± 0.19
C16:0	2.73 ± 0.53
C16:1	0.74 ± 0.06
iso-C17:0	0.70 ± 0.03
anteiso-C17:0	8.22 ± 0.32
iso-C18:0	5.28 ± 0.44
C18:0	1.45 ± 0.10
C18:1	0.66 ± 0.08
anteiso-C19:0	6.00 ± 0.58
iso-C20:0	2.14 ± 0.31
anteiso-C21:0	1.83 ± 0.33
iso-BCFA	44.62 ± 0.50
anteiso-BCFA	40.20 ± 0.89
Total BCFA	84.82 ± 1.22
Others[1]	15.18 ± 1.22

[1]Others = straight-chain fatty acids.

(**LDLC**), and high-density lipoprotein cholesterol were analyzed by a fully automatic biochemical analyzer (BS-2000M2, Mindray, Shenzhen, China). The nonesterified fatty acid (**NEFA**) level in plasma was determined by following the manufacturer's instructions for commercial colorimetric analysis kits (Nanjing Jiancheng Institute of Bioengineering, Nanjing, China). Plasma insulin, haptoglobin (**HP**), serum amyloid A (**SAA**), IL-1β, IL-6, IL-8, and IL-10 concentrations were determined using commercial bovine ELISA kits (Jiangsu Jingmei Biotechnology Co. Ltd., Yancheng, China).

Fecal SCFA Concentrations. Short-chain fatty acids, including acetic acid, propionic acid, isobutyric acid, butyric acid, isovaleric acid, and valeric acid, were quantified in fecal samples by GC. Briefly, 1 g of fecal samples was weighed, dissolved, homogenized, and then centrifuged at 12,000 × *g* for 15 min at 4°C. Then the supernatant, mixed with crotonic acid as internal standard, was filtered through a 0.22-mm sterile membrane and stored in 2-mL screw-cap vials before SCFA analysis using a GC system (Shimadzu, GC-2010, Japan; Kodithuwakku et al., 2021).

Small Intestine Histomorphology. The fixed intestinal tissue was processed and embedded in paraffin. Then the samples were fixed on a paraffin microtome (Junjie Electronics Co. Ltd., Wuhan, China) and two 4-μm-thick tissue sections were cut. The tissue sections were placed on slides and stained with hematoxylin and eosin (**H&E**) staining solution. The prepared sections were analyzed using the pathological section scanner (EFL, Suzhou, China). The ZYFviewer (2.0.1.10, Zhiying Medical Technology Co. Ltd., Jinan, China) pathological image analysis system was used to measure the villus height (from the tip of the villus to the villus:crypt interface) and crypt depth (from the villus:crypt interface to the muscularis mucosae; van Keulen et al., 2020).

Quantitative Real-Time PCR. Frozen duodenum, jejunum, and ileum tissues were separately ground completely in liquid nitrogen using a mortar and pestle. Tissue RNA was extracted using the TriZol (Invitrogen, Waltham, MA) method described by Wang et al. (2023), and gDNA Eraser (TaKaRa, Baori Medical Technology Co. Ltd., Beijing, China) was used to remove genomic DNA. The RT Primer Mix and TB Green Premix Ex *Taq*II (TaKaRa, Baori Medical Technology Co. Ltd., Beijing, China) were used for cDNA synthesis, and quantitative PCR determination of various genes was performed in the laboratory. All primers used in the assay were synthesized by Primer Premier 5.0 (Canada), and their double-ended sequences are shown in Supplemental Material S1 (see Notes). The qPCR amplification and detection were conducted using the 7900HT rapid real-time PCR system (Applied Biosystems, Foster City, CA). The relative mRNA expression was determined by the $2^{-\Delta\Delta Ct}$ method (Gudelska et al., 2020).

High-Throughput Amplicon Sequencing of 16S rRNA Gene and Data Analysis

At 14 and 28 d of age, 10 fecal samples from the BCFA group and control group were randomly selected, then analyzed using high-throughput sequencing at Lianchuan Biotechnology (Hangzhou, China). Total microbial genomic DNA was extracted from each fecal sample using the E.Z.N.A.Stool DNA Kit (D4015, Omega Inc., Norwalk, CT). The purity and integrity of DNA were detected by using a NanoDrop microspectrophotometer (Thermo Fisher Scientific, Waltham, MA) and agarose gel electrophoresis, respectively. Primers 341F (5′-CCTACGGGNGGCWGCA-3′) and 805R (5′-GACTACHVGGGTATCTAATCC-3′) were used to amplify the V3-V4 region of the prokaryotic (bacterial and archaeal) 16S rRNA gene (Liang et al., 2023). The PCR products were purified using AMPure XT microbeads (Beckman Coulter Genomics, Danvers, MA) and quantified using Qubits (Invitrogen). Sequencing was performed according to NovaSeq's PE250 mode pooling. Amplicon sequencing data were analyzed using Quantitative Insights Into Microbial Ecology 2 (QIIME2, version 2022.08; Bolyen

Table 3. Effect of BCFA-DFL supplementation on growth performance of dairy calves

	Treatment[1]			*P*-value[2]						
Item	Control	BCFA	SEM	Treatment	Time	Sex	Trt × time	Trt × sex	Time × sex	Trt × time × sex
BW, kg										
d 14	47.28	50.02	0.870	0.04	—	0.29	—	0.36	—	—
d 28	56.53	61.85	1.386	0.11	—	0.10	—	0.59	—	—
Overall	51.90	55.93	0.975	0.02	<0.01	0.10	0.16	0.40	0.75	0.77
ADG, kg/d										
d 0–14	0.38	0.58	0.066	<0.01	—	0.31	—	0.43	—	—
d 14–28	0.66	0.84	0.087	0.16	—	0.20	—	0.64	—	—
Overall	0.52	0.71	0.053	0.01	<0.01	0.09	0.99	0.46	0.60	0.96
Whole milk DMI, g/d										
d 0–14	722.77	728.35	6.523	0.55	—	0.01	—	0.99	—	—
d 14–28	903.24	973.98	15.607	<0.01	—	0.74	—	0.74	—	—
Overall	813.01	851.17	7.847	0.01	<0.01	0.41	0.02	0.74	0.20	0.77
Starter DMI, g/d										
d 3~14	26.47	39.42	13.295	0.50	—	0.18	—	0.73	—	—
d 14~28	112.35	106.80	31.067	0.90	—	0.27	—	0.93	—	—
Overall	69.41	73.11	18.494	0.89	<0.01	0.16	0.67	0.96	0.60	0.81
Total DMI, g/d										
d 0–14	743.57	759.33	13.44	0.42	—	0.02	—	0.80	—	—
d 14–28	1,015.59	1,080.78	38.91	0.25	—	0.46	—	0.95	—	—
Overall	879.58	920.05	21.765	0.20	<0.01	0.16	0.38	0.89	0.93	0.98
Feed efficiency,[3] kg/kg										
d 0–14	0.51	0.77	0.071	0.02	—	0.06	—	0.45	—	—
d 14–28	0.64	0.79	0.094	0.29	—	0.73	—	0.66	—	—
Overall	0.58	0.78	0.056	0.02	0.38	0.14	0.52	0.40	0.39	0.92
Withers height, cm										
d 0–14	82.10	82.21	0.713	0.22	—	0.23	—	0.16	—	—
d 14–28	84.31	86.69	0.841	0.33	—	0.71	—	0.55	—	—
Overall	83.17	84.50	0.633	0.09	<0.01	0.16	0.11	0.13	0.71	0.62
Body length, cm										
d 0–14	72.73	75.76	0.816	0.84	—	0.72	—	0.41	—	—
d 14–28	77.06	80.96	0.903	0.52	—	0.98	—	0.80	—	—
Overall	74.89	78.36	0.596	0.72	<0.01	0.78	0.57	0.45	0.98	0.36
Heart girth, cm										
d 0–14	86.43	86.83	0.466	0.15	—	<0.01	—	0.55	—	—
d 14–28	89.66	93.20	1.415	0.89	—	0.06	—	0.76	—	—
Overall	87.98	89.96	0.759	0.03	<0.01	0.01	0.11	0.73	0.58	0.72
Hip height, cm										
d 0–14	79.55	79.74	0.847	0.83	—	0.33	—	0.15	—	—
d 14–28	82.59	84.83	0.909	0.10	—	0.73	—	0.62	—	—
Overall	81.11	82.25	0.741	0.14	<0.01	0.30	0.27	0.64	0.74	0.17

[1]Control = calves fed whole milk containing 0.4 g of emulsifier (n = 12; 6 males, 6 females); BCFA = calves fed whole milk containing 4 g of BCFA-DFL (n = 12; 6 males, 6 females); all the treatments were carried out during the morning feeding.

[2]Trt = treatment.

[3]Feed efficiency was calculated as the ratio of ADG (kg/d) to total DMI (kg/d).

et al., 2019). Raw reads were filtered for high quality, assembled, and chimeric sequences were removed using DATA2, resulting in unique amplicon sequence variants (**ASV**; Callahan et al., 2016). We used the SILVA database classifier (version 138, https://www.arb-silva.de/documentation/release-138/) to classify ASV based on 99% sequence similarity. In the study of fecal microorganisms, a genus can be defined as present if it can be identified in more than 50% of the samples from each group (Ma et al., 2020; Xin et al., 2021). Determinations of α and β diversities were also conducted in QIIME 2. All sequences were deposited in the National Center for Biotechnology Information Sequence Read Archive (https://www.ncbi.nlm.nih.gov/sra) under the project number PRJNA1222321.

Statistical Analysis

The sample size calculation for this study was performed using the PROC POWER procedure in SAS (version 9.4; SAS Institute Inc., Cary, NC), based on the method described by Ansari et al. (2022). The assessment was carried out for the following primary response variables: weight gain (Górka et al., 2011; Lu et al., 2022), diarrhea frequency (Ma et al., 2024), and villi height (Pyo et al., 2020). The selection of these variables was

Table 4. Effect of BCFA-DFL supplementation on total-tract apparent digestibility of nutrients in dairy calves

	Treatment[1]			*P*-value[2]		
Digestibility, %	Control	BCFA	SEM	Treatment	Sex	Trt × sex
DM	70.21	75.66	0.864	<0.01	0.13	0.38
NDF	54.32	61.25	1.440	<0.01	0.47	0.75
ADF	28.79	40.47	2.000	<0.01	0.68	0.47
CP	73.78	80.14	0.838	<0.01	0.99	0.54

[1]Control = calves fed whole milk containing 0.4 g of emulsifier (n = 12; 6 males, 6 females); BCFA = calves fed whole milk containing 4 g of BCFA-DFL (n = 12; 6 males, 6 females); all the treatments were carried out during the morning feeding.

[2]Trt = treatment.

informed by recent published literature. The predicted sample sizes were 12 and 4 calves per treatment group for variables related to growth performance and intestine development, respectively ($\alpha = 0.05$ and power = 0.85). At the end of the experiment, all the male calves (n = 6 per group) were slaughtered.

All data were checked for normality and outliers using the UNIVARIATE procedure in SAS (version 9.4; SAS Institute Inc.) before any statistical analyses were conducted. Data of BW, ADG, DMI, body measurements, fecal VFA, and plasma metabolites were analyzed using the PROC MIXED procedure in SAS. The model included block as a random effect, treatment, sex, time, treatment × sex, treatment × time, sex × time, and treatment × time × sex as fixed effects. Initial BW and body measurements were used as covariates for the growth performance data of the repeated measures. Various covariance structures, including simple covariance structure, compound symmetry, first-order autoregressive, autoregressive heterogeneous, and unstructured, were evaluated and selected based on the minimum corrected Akaike information criterion. All models were visually assessed for fit and residual uniformity, and covariates were assessed for collinearity. The normality of the residuals was tested via PROC UNIVARIATE in SAS. The total-tract apparent digestibility of nutrients was analyzed with the PROC MIXED procedure, incorporating block as a random effect and treatment, sex, and treatment × sex as fixed effects.

As described by Ansari et al. (2022) and Nowroozinia et al. (2022), data on elevated rectal temperature (≥39.4°C), diarrhea (fecal score ≥3), and medication use were analyzed using logistic regression with a binomial distribution in the PROC GLIMMIX procedure, and the odds ratio (**OR**) was used to compare the likelihood of these events between treatment groups. Analysis using a Poisson distribution within the PROC GENMOD procedure was applied to the number of days with elevated rectal temperature, diarrhea frequency, diarrhea duration, and medication administration.

Histomorphology parameters and gene expression of the intestine were analyzed using PROC MIXED procedure. Treatment and block were considered as fixed and random effects, respectively. Gene expression of the intestinal tissue was represented as means ± SD and displayed using the GraphPad Prism 10.0 program (GraphPad Software, San Diego, CA), where $*0.01 < P \leq 0.05$, $**0.001 < P \leq 0.01$, and $*** P \leq 0.001$. The Tukey–Kramer adjustment was applied to account for multiple comparisons. The relative abundance and α diversity of fecal microbiota were analyzed by the Wilcoxon test in R software (version 4.1.2; https://www.r-project.org). Significance was declared at $P \leq 0.05$ and trends at $0.05 < P \leq 0.10$.

RESULTS

Fatty Acid Composition of BCFA-DFL

The BCFA-DFL components extracted in this study are shown in Table 2. The carbon chain length of fatty acids was concentrated in the range of C10 to C21. The content of total BCFA in the product components was up to 84.82%, of which the contents of iso-BCFA and anteiso-BCFA were 44.61% and 40.20%, respectively. The highest component among BCFA-DFL was anteiso-C15:0 (16.56%), followed by iso-C14:0 (14.47%), iso-C16:0 (14.27%), and anteiso-C17:0 (8.22%). The BCFA-DFL also contained approximately 15% straight-chain fatty acids, of which C14:0 (5.49%) was the most abundant, followed by C16:0 (2.73%). In addition, 2 UFA, C16:1 (0.74%) and C18:1 (0.66%), were also detected, but their contents were less than 1%.

Growth Performance, Total-Tract Apparent Digestibility, and Diarrhea Frequency

Supplementation with BCFA-DFL significantly increased the BW, ADG, feed efficiency, and heart girth of calves during the 0- to 28-d period (Table 3; $P < 0.05$). Specifically, calves supplemented with BCFA-DFL exhibited higher ADG and feed efficiency during the 0- to 14-d period, as well as higher BW on d 14 ($P < 0.05$).

Table 5. Logistic model for elevated rectal temperature (≥39.4°C), occurrence of diarrhea (fecal score ≥ 3), and medication occurrence during the trial (28 d) as influenced by feeding whole milk without (control) or with BCFA-DFL (BCFA; 4 g/d) to dairy calves

Variable and comparison	Coefficient	SEM	OR[1]	95% CI	*P*-value
Rectal temperature					
Control vs. BCFA	0.5066	0.576	1.66	0.535, 5.144	0.38
Female vs. male	−0.1480	0.572	0.86	0.281, 2.650	0.80
Diarrhea occurrence					
Control vs. BCFA	1.0517	0.299	2.86	1.591, 5.149	<0.01
Female vs. male	−0.2323	0.299	0.79	0.441, 1.426	0.44
Diarrhea medication occurrence					
Control vs. BCFA	1.0566	0.426	2.88	1.246, 6.642	0.01
Female vs. male	−0.0571	0.426	0.945	0.409, 2.180	0.89

[1]The odds ratio (OR) indicates the probability of having elevated rectal temperature (≥39.4°C), diarrhea (fecal score ≥ 3), or needing medication for the experimental treatments. If the OR is >1, a given treatment in the comparison is more likely to have a greater occurrence of diarrhea than another treatment. If the OR is <1, it indicates that a given treatment has a lower probability of occurrence than another treatment.

Throughout the experiment, no significant differences were observed in starter DMI or total DMI between the 2 groups ($P > 0.05$). Additionally, calf body length and hip height were not affected by BCFA supplementation during the trial period ($P > 0.05$). Although BCFA-DFL supplementation increased whole milk DMI during the 14- to 28-d and 0- to 28-d periods ($P < 0.05$), it had no significant effect on starter DMI or total DMI. Compared with the control group, calves in the BCFA-treated group tended to have increased withers height ($P = 0.09$). Furthermore, the total-tract apparent digestibility of DM, NDF, ADF, and CP was significantly higher in calves supplemented with BCFA ($P < 0.01$; Table 4).

The logistic model detailing elevated rectal temperature, incidence of diarrhea, and frequency of medication treatment in calves is summarized in Table 5. Compared with the BCFA group, calves in the control group exhibited a higher incidence of diarrhea (OR = 2.86; 95% CI = 1.591–5.149; SEM = 0.299; $P < 0.01$). The likelihood of drug treatment necessitated by diarrhea was also greater in the control group (OR = 2.88; 95% CI = 1.246–6.642; SEM = 0.426; $P = 0.01$). No significant difference was observed in the frequency of increased rectal temperature between the 2 treatment groups (OR = 1.66; $P = 0.38$). Additionally, no significant differences were observed between female and male calves regarding elevated rectal temperature ($P = 0.80$), diarrhea ($P = 0.38$), or the likelihood of drug treatment due to diarrhea ($P = 0.89$). The Poisson regression results regarding the duration of elevated rectal temperature, the incidence and frequency of diarrhea, and the number of medication administrations during the trial are summarized in Table 6. No significant difference was observed in the number of days with elevated rectal temperature between the treatment groups ($P = 0.26$). The control group exhibited a longer duration ($P = 0.01$), a higher frequency ($P < 0.01$), and an increased number of days requiring drug treatment ($P < 0.01$) for diarrhea compared with the BCFA group.

Plasma Metabolites

Plasma data for biochemical parameters and inflammatory cytokines measured on d 14 and 28 for all enrolled calves are presented in Table 7. Feeding BCFA-DFL significantly reduced cholesterol levels ($P = 0.04$) and HP levels ($P = 0.02$) in the serum of calves. Compared with the control group, the plasma levels of LDLC and IL-1β in the BCFA group exhibited a declining trend ($P = 0.09$). The levels of other blood parameters measured did not differ significantly between treatments ($P > 0.10$).

Small Intestine Histomorphology

Histomorphology of the small intestine is shown in Figure 2. The measured villus and crypt data for duodenum, jejunum, and ileum were presented in Table 8. There were no significant differences in villus height, crypt depth, or the villus:crypt ratio in the duodenum, jejunum, and ileum between the 2 groups ($P > 0.05$). Calves in the BCFA group showed a downward trend in crypt depth and an upward trend in villus:crypt ratio in the ileal tissue compared with the control group ($P = 0.08$).

Quantitative Real-Time PCR

The effects of BCFA-DFL on intestinal gene expression in calves are presented in Figure 3. Compared with the control group, calves supplemented with BCFA-DFL exhibited significant upregulation of key tight junction proteins. Specifically, the expression of *TJP1* (encoding ZO-1) was increased in the duodenum and ileum ($P < 0.05$), and *CLDN2* expression was higher in the duodenum and jejunum ($P < 0.05$). Furthermore, calves fed

Table 6. Poisson regression for days with elevated rectal temperature (≥39.4°C), frequency and duration of diarrhea (fecal score ≥3), and medication occurrence during the experiment (d 1–28) as influenced by feed whole milk without (control) or with BCFA-DFL (BCFA; 4 g/d) to dairy calves

Item[2]	Treatment[1]			*P*-value		
	Control	BCFA	SEM	Treatment	Sex	Trt × sex
Days with elevated rectal temperature (≥39.4°C)	1.83	1.25	0.513	0.26	0.64	0.92
Duration of diarrhea, d/calf	6.42	2.75	0.214	0.01	0.32	0.10
Diarrhea frequency, %	22.92	9.82	0.232	<0.01	0.36	0.17
Diarrhea medication, d	3.08	1.17	0.316	<0.01	0.84	0.47

[1]Control = calves fed whole milk containing 0.4 g of emulsifier (n = 12; 6 males, 6 females); BCFA = calves fed whole milk containing 4 g of BCFA-DFL (n = 12; 6 males, 6 females); all the treatments were carried out during the morning feeding.

[2]Duration of diarrhea is average length of diarrhea (no. of days per calf); diarrhea frequency (%) = (number of calves with diarrhea × number of diarrhea days)/(total number of calves × number of test days) × 100. Days with elevated rectal temperature is the average length of fever (no. of days per calf).

BCFA-DFL had higher gene expression levels of *DEFB1* and *DEFB4* in intestinal tissue ($P < 0.05$). The expression of *REG3α* and *REG3γ* in the jejunum of the BCFA group were higher than the control group ($P < 0.01$), and the *REG3α* in the duodenum showed a similar varying tendency in the 2 treatment groups ($P < 0.01$). Feeding BCFA-DFL significantly increased *IL-10* gene expression levels in the duodenum, jejunum, and ileum tissues of calves ($P < 0.05$). The mRNA expression levels of *IL-1β* in the duodenum and *IL-1β* and *TNF-α* in the jejunum of the BCFA group were significantly lower than those of the control group ($P < 0.05$). The expressions of all other genes analyzed were unaffected by treatment.

Fecal Fermentation Parameters

Compared with the control group, calves fed BCFA-DFL exhibited elevated concentrations of total SCFA, isobutyrate, and butyrate (Table 9; $P < 0.05$). A treatment × time interaction was observed for fecal butyrate ($P = 0.01$) and isovalerate ($P < 0.01$) on d 14 and 28. On d 14, the butyrate concentration in the feces of calves in the BCFA group was significantly higher than that of the control group ($P < 0.05$), and levels in both treatment groups were comparable on d 28 ($P > 0.05$). Furthermore, isovalerate concentrations were significantly greater in the feces of calves fed BCFA-DFL compared with the control group on d 28 ($P < 0.05$), although no differences between treatments were noted on d 14 ($P > 0.05$). The administration of BCFA-DFL did not affect the concentrations of acetate, propionate, and valerate.

Fecal Bacterial Communities of Calves

In the study of fecal microorganisms, a genus can be defined as present if it can be identified in more than 50% of the samples from each group. Fecal microbial structure, diversity indexes, and differences between calves in the 2 treatment groups at 14 and 28 d of age are presented in Figure 4 and Table 10. The composition of fecal microbiota at 14 and 28 d of age was different between the groups at the phylum and genus levels (Figure 4a, b). As shown in Figure 4c–f, α diversity analysis of the 2 groups in the feces of 14-d-old calves showed that the community contained the Chao1 index, the Shannon index, the Simpson index, and the Pielou_e index, and significant differences were observed unless Chao1 index ($P < 0.05$). In addition, the β diversity analysis demonstrated distinct differences between the 2 treatment groups (Figure 4g). Feeding BCFA-DFL did not affect fecal microbiota α and β diversity results in 28-d-old calves ($P > 0.05$).

As demonstrated in Figure 4b and Table 10, *Collinsella*, *Lactobacillus*, and *HT002* were the predominant genera in calf feces on d 14 and 28. The relative abundance of *Subdoligranulum* in the feces of calves feeding BCFA was higher at 14 d of age ($P < 0.01$). It remained the dominant genus in the fecal microbiota over time, but its relative abundance tended to be consistent with that of the control group at 28 d of age ($P = 0.68$). The relative abundance of *Faecalibacterium*, *Clostridia_UCG-014_unclassified*, *Bacteroides*, *Sharpea*, *Butyricicoccus*, and *Fournierella* differed between treatments only at 14 d of age, with higher abundance in the feces of calves fed BCFA-DFL ($P < 0.05$). In the feces of 14-d-old calves, *Peptostreptococcus* and *Fusobacterium* were present only in the control group, and *Prevotella_9* and *Paludicola* were present only in the BCFA group. Furthermore, the relative abundance of *Pseudoflavonifractor* was higher in the feces of calves in the BCFA group than in the control group at both d 14 and d 28 ($P < 0.05$). The relative abundance of *Oscillospiraceae_UCG-005*, *Ruminococcus*, *Eubacterium_coprostanoligenes_group_unclassified*, and *Christensenellaceae_R-7_group* differed between treatments only at 28 d of age, with higher abundance in the feces of calves fed BCFA-DFL ($P < 0.05$). In the feces of 28-d-old calves, *Megasphaera* was detected only

Table 7. Effect of BCFA-DFL supplementation on plasma metabolites in dairy calves

Item[3]	Treatment[1]			*P*-value[2]						
	Control	BCFA	SEM	Treatment	Time	Sex	Trt × time	Trt × sex	Time × sex	Trt × time × sex
Biochemical parameter										
TG, mmol/L	0.40	0.33	0.043	0.27	<0.01	0.70	0.16	0.71	0.54	0.43
NEFA, mmol/L	0.25	0.28	0.017	0.22	0.42	0.58	0.17	0.57	0.97	0.79
Cholesterol, mmol/L	3.33	2.80	0.169	0.04	0.03	0.17	0.18	0.98	0.23	0.46
HDLC, mmol/L	2.34	2.55	0.147	0.32	0.40	0.20	0.83	0.91	0.46	0.58
LDLC, mmol/L	0.75	0.56	0.077	0.09	0.01	0.32	0.83	0.27	0.39	0.85
Total protein, g/L	63.78	65.58	1.203	0.30	0.01	0.97	0.33	0.23	0.61	0.83
ALB, g/L	30.42	31.67	0.530	0.11	0.01	0.88	0.33	0.89	0.70	0.87
Insulin, mIU/L	13.96	14.40	0.590	0.60	0.18	0.85	0.10	0.60	0.91	0.79
HP, μg/mL	35.08	32.78	0.644	0.02	0.12	0.09	0.09	0.66	0.75	0.75
SAA, μg/L	439.38	442.21	18.200	0.91	0.01	0.76	0.71	0.68	0.59	0.31
Inflammatory cytokine, ng/L										
IL-1β	44.52	41.84	1.077	0.09	0.07	0.15	0.17	0.36	0.43	0.44
IL-6	19.76	18.61	0.688	0.25	<0.01	0.88	0.40	0.92	0.08	0.80
IL-8	422.70	420.00	23.671	0.94	0.05	0.16	0.95	0.97	0.33	0.38
IL-10	39.73	40.71	1.475	0.64	0.02	0.25	0.83	0.68	0.62	0.43

[1]Control = calves fed whole milk containing 0.4 g of emulsifier (n = 12; 6 males, 6 females); BCFA = calves fed whole milk containing 4 g of BCFA-DFL (n = 12; 6 males, 6 females); all the treatments were carried out during the morning feeding.

[2]Trt = treatment.

[3]TG = triglyceride; NEFA = nonesterified fatty acids; HDLC = high-density lipoprotein cholesterol; LDLC = low-density lipoprotein cholesterol; ALB = albumin; HP = haptoglobin; SAA = serum amyloid A.

in the control group, and *Terrisporobacter* was present only in the BCFA group.

DISCUSSION

Growth Performance and Total-Tract Apparent Digestibility

This present research revealed that supplementation with BCFA-DFL increased ADG and body measurements of calves, indicating a positive influence on the growth performance. Notably, the total DMI of calves remained unaffected by the treatment, and calves fed BCFA-DFL demonstrated a higher feeding efficiency. Previous studies have shown that BCFA do not merely function as an energy source but can also be transported to tissues and organs for absorption and metabolism (Ran-Ressler et al., 2011; Liu et al., 2017). It has been reported that iso-C17:0 can act as a chemical or nutritional factor to promote embryo development (Kniazeva et al., 2008). Additionally, both iso-C17:0 and iso-C15:0 are essential fatty acids for promoting the growth and development of *Caenorhabditis elegans* (Kniazeva et al., 2004; Jia et al., 2016). From this, we speculate that BCFA may be able to act as a nutritional factor to promote the growth and development of calves. Additionally, BCFA can prevent and alleviate intestinal inflammation by regulating intestinal immunity and the structure of microorganisms (Ran-Ressler et al., 2011; Zhang et al., 2024). During this stage, when calves are vulnerable to diseases such as diarrhea, their health status becomes a key factor influencing growth performance (Schinwald et al., 2022; Moreira et al., 2026). Chang et al. (2022) demonstrated that supplementing newborn calves with milk oligosaccharides reduced the frequency of diarrhea during the d 0 to 28 period and significantly increased the ADG. The magnitude of this improvement in ADG was comparable to the difference observed between the 2 treatment groups in the present study. Moreover, Schinwald et al. (2022) found that the differences in ADG caused by diarrhea could be observed at 14 d of age. Similar results were also produced in this study. This might be related to the relatively large abundance difference of microorganisms such as *Subdoligranulum* and *Faecalibacterium* that produce butyric acid and have anti-inflammatory functions between the 2 treatment groups at 14 d of age (Oikonomou et al., 2013; Xin et al., 2021). The butyric acid serves as an energy source for intestinal cells and can protect the intestines (Hodgkinson et al., 2023). Consequently, the superior intestinal health in the BCFA group calves can likely be attributed to elevated butyric acid concentrations in their feces. Furthermore, research reported a positive correlation between the relative abundance of *Faecalibacterium* in the feces of calves and their growth performance (Hennessy et al., 2021). Multiple studies have shown that when calves develop diarrhea, infection or inflammatory responses lead to reduced intake of liquid feed, often accompanied by impaired digestion and malabsorption (Knauer et al., 2017; Conboy et al., 2022; Bierlein and Gross, 2026). The DMI of whole milk was lower in calves of the control group, which was likewise attributed to feed refusal

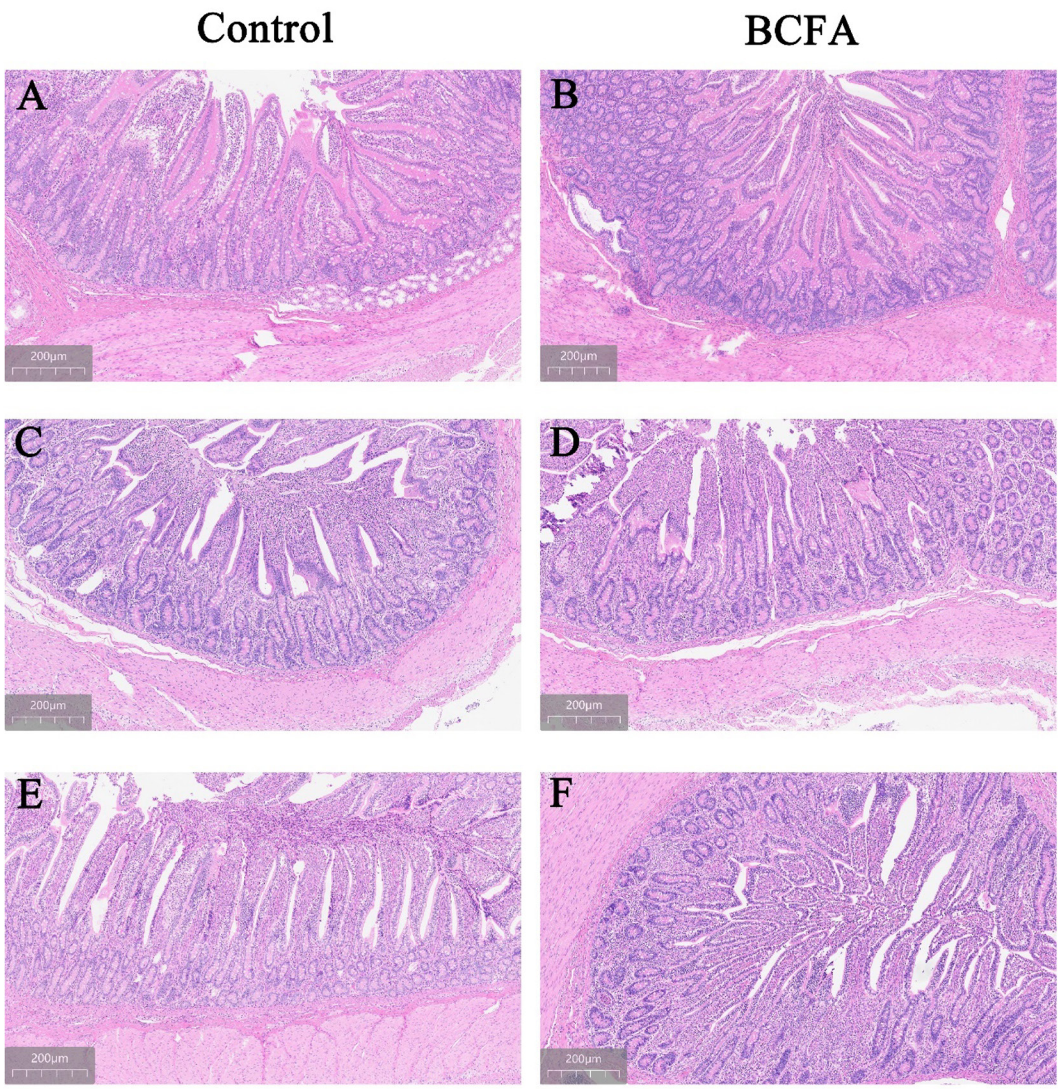

Figure 2. Light micrographs of the duodenum (A–B), jejunum (C–D), and ileum (E–F) of 28-d-old calves (H&E staining, 4-μm sections, 10× magnification, gray scale bars represent 200 μm) fed unsupplemented whole milk or milk supplemented with BCFA-DFL.

behavior among calves experiencing diarrhea during the d 14 to 28 period. Preweaning ADG has been reported to be positively correlated with whole milk consumption (Johnson et al., 2018). Therefore, the differences in the growth performance of calves aged 14 to 28 d are also related to the changes in raw milk intake caused by diarrhea. In addition, a healthy gastrointestinal state in calves is more conducive to the emptying of feed (Xu et al., 2022). The BCFA-DFL did not affect the feed intake of starter in calves but increased the apparent digestibility of nutrient components. The digestive and absorptive capacity of the gut is largely determined by the integrity of the gut morphology and barrier, moreover, *ZO-1* was a crucial factor in maintaining the integrity of the intestinal barrier (Shuai et al., 2023). Therefore, we speculate that the higher nutrient digestibility in the calves of the BCFA group is related to the higher expression level of *ZO-1* in their small intestines.

Plasma Metabolites

High levels of cholesterol and LDLC in human blood are considered to pose a risk of causing metabolic diseases, such as atherosclerosis (Yaghi and Elkind, 2015). Numerous BCFA, including iso-C15:0, anteiso-C15:0, iso-C16:0, iso-C17:0, anteiso-C17:0, and iso-C18:0, have long been recognized as being closely linked to obesity and metabolic inflammation; moreover, within tissues or organs, the concentrations of these BCFA exhibit a negative correlation with cholesterol levels (Goz-

Table 8. Histomorphology of the small intestine (duodenum, jejunum, and ileum) of 28-d-old calves fed unsupplemented whole milk or milk supplemented with BCFA-DFL

	Treatment[1]			
Item[2]	Control	BCFA	SEM	*P*-value
Duodenum				
Villus height, μm	584	608	14.176	0.43
Crypt depth, μm	346	343	18.925	0.93
Villus height:crypt depth	1.82	1.86	0.074	0.77
Jejunum				
Villus height, μm	550	577	16.718	0.44
Crypt depth, μm	344	358	13.803	0.63
Villus height:crypt depth	1.67	1.68	0.065	0.90
Ileum				
Villus height, μm	557	560	8.921	0.88
Crypt depth, μm	405	370	10.355	0.08
Villus height:crypt depth	1.40	1.54	0.041	0.08

[1]Control = calves fed whole milk containing 0.4 g of emulsifier (n = 6; 6 males); BCFA = calves fed whole milk containing 4 g of BCFA-DFL (n = 6; 6 males); all the treatments were carried out during the morning feeding.

[2]Duodenum, jejunum, and ileum samples were collected from the intestines of male calves slaughtered at 29 d of age.

dzik et al., 2023). In this study, the cholesterol levels in the plasma of calves fed BCFA-DFL were significantly lower compared with those of the control group calves, whereas the levels were within the range reported in published research (Marcato et al., 2024; Wilms et al., 2024). However, it is worth noting that blood cholesterol is also a significant indicator of the health status of calves, increasing with age and showing a positive correlation with the survival rate of calves (Goetz, et al., 2021; Marcato, et al., 2024). This is contrary to our results. Therefore, we hypothesize that BCFA, while regulating intestinal health, also play a role in reducing cholesterol levels in the blood. This change may have little to do with the health status of the calves. A recent study has shown that BCFA extracted from yak butter can inhibit the synthesis of plasma cholesterol in mice by regulating the expression of the HMGCR gene while also reducing plasma LDLC levels (Tan et al., 2023). This finding may account for the decrease in plasma cholesterol levels observed in the calves fed BCFA-DFL in the present study. However, the pathways by which BCFA regulate LDLC in plasma remain unknown. Additionally, Mickiewicz et al. (2024) used lipoprotein separation therapy to decrease the LDLC content in the blood of atherosclerotic patients, and as a result the levels of total BCFA and anteiso-BCFA in the blood were increased (Mickiewicz et al., 2024). Therefore, we can conclude that there is a negative correlation between BCFA and LDLC to a certain extent, and feeding BCFA-DFL can reduce the LDLC level in the plasma of calves.

Calves have a higher sensitivity to inflammatory stimuli in the early stages of life (Lopes et al., 2021).

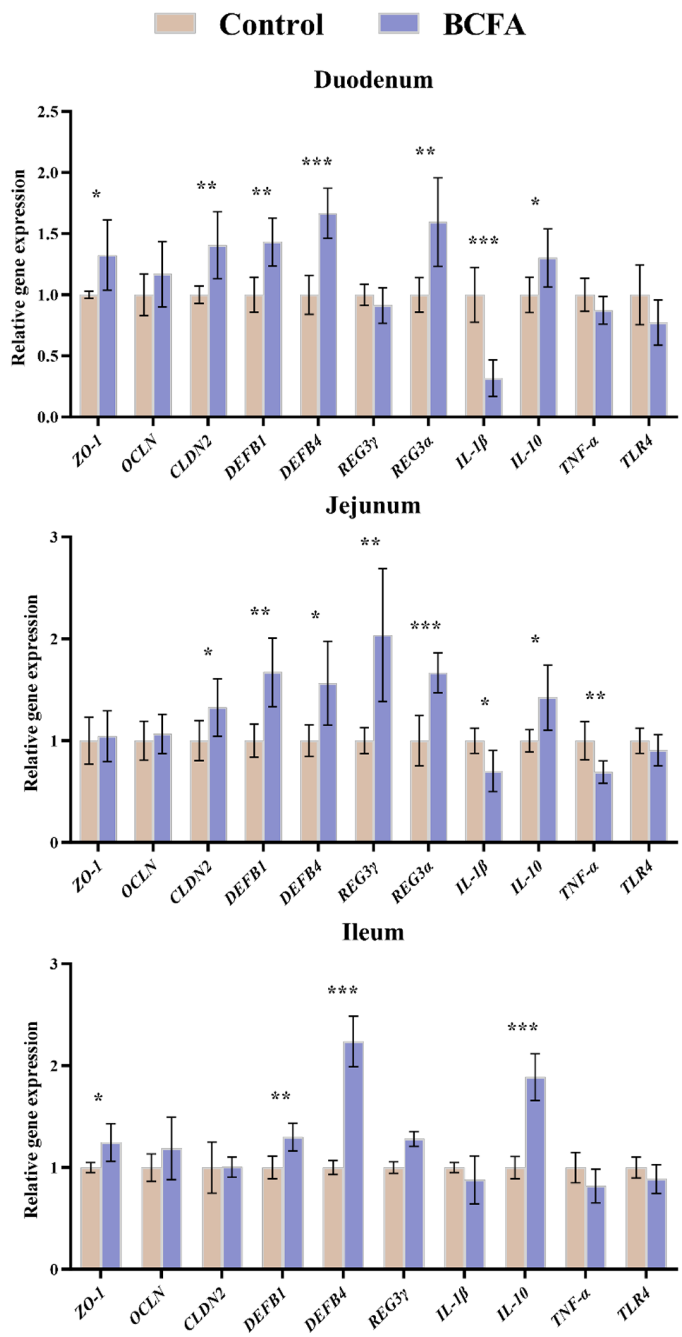

Figure 3. Effect of BCFA-DFL supplementation on the relative mRNA expression ratio of genes in the duodenum, jejunum and ileum of calves. $*P \leq 0.05$, $**P \leq 0.01$, $***P \leq 0.001$. Error bars represent SD. ZO-1 = tight junction protein 1; OCLN = occludin; CLDN2 = claudin-2; DEFB1 = β-defensin 1; DEFB4 = β-defensin 4; REG3α = regenerating islet-derived 3α; REG3γ = regenerating islet-derived 3γ; IL-1β = interleukin 1β; IL-10 = interleukin 10; TNF-α = tumor necrosis factor α; TLR4 = toll-like receptors 4.

Plasma HP and IL-1β are mediators of the immune response to stress and infection in calves (Jaramillo et al., 2022; Caliskan et al., 2023; Fu et al., 2024), and have

Table 9. Effect of BCFA-DFL supplementation on feces SCFA in dairy calves

Item	14 d		28 d		SEM	P-value[1]						
	Control	BCFA	Control	BCFA		Treatment	Time	Sex	Trt × time	Trt × sex	Time × sex	Trt × time × sex
Total SCFA, μmol/g	68.82	74.63	101.33	133.09	7.505	0.02	<0.01	0.85	0.11	0.71	0.89	0.77
Individual SCFA, %												
Acetate	65.57	57.02	60.70	60.59	2.617	0.13	0.80	0.41	0.11	0.35	0.38	0.92
Propionate	20.39	21.14	25.41	19.45	1.879	0.13	0.44	0.65	0.13	0.15	0.72	0.96
Isobutyrate	1.54	1.52	1.79	3.54	0.415	0.02	0.03	0.38	0.08	0.33	0.97	0.40
Butyrate	9.48[b]	18.52[a]	8.03[b]	10.06[b]	1.307	<0.01	<0.01	0.11	0.01	0.44	0.03	0.54
Isovalerate	2.87[b]	1.46[b]	2.46[b]	4.61[a]	0.515	0.44	0.02	0.49	<0.01	0.15	0.78	0.16
Valerate	0.14	0.34	1.62	1.76	0.179	0.31	<0.01	0.51	0.88	0.16	0.71	0.58

[a,b]Different letters in a row indicate significant differences of treatment effect ($P < 0.05$).

[1]Trt = treatment.

been reported to be associated with calf morbidity and mortality (Chitko-McKown et al., 2021; Gao et al., 2025). The plasma levels of HP and IL-1β were lower in the BCFA-DFL fed calves in the present study than in control calves. Similarly, feeding mice with BCFA in goat milk reduced plasma levels of IL-1β in mice with colitis (Zhang et al., 2024). Moreover, the prevalence of diarrhea was lower in the BCFA group, and some calves never developed diarrhea during the experiment. Therefore, we believe that the degree of stress in the 2 groups of calves was different, so it is reasonable that HP and IL-1β would have differed between treatments. Overall, the effects of BCFA-DFL on blood metabolism and inflammatory status in calves were positively affected.

Intestinal Development and Immunity

In the present study, BCFA-DFL exerted no significant effect on the morphological parameters of the small intestine in calves. It merely exhibited a tendency to enhance the ratio of villus height to crypt depth in the ileum. According to research insights, the villus height and crypt depth within the intestine serve as indicators capable of reflecting the potential maturity and absorption capacity of intestinal epithelial cells, as well as the permeability of the intestinal tract (Abdelsattar et al., 2022). Similarly, in other published research, altering the fatty acid content and structure of liquid diets could increase the expression levels of proteins related to intestinal mucosal integrity in the small intestine of calves, such as *ZO-1*, but had no effect on intestinal villus height and crypt depth (Mellors et al., 2023; Welboren et al., 2023). In this experiment, BCFA-DFL promoted the expression of the tight junction genes *ZO-1* and *CLDN2* in the small intestine of 28-d-old calves. *ZO-1* and *CLDN2* can protect the intestinal mucosa from potentially toxic substances, thereby maintaining the intestinal barrier function (Xie et al., 2023). In addition, previous studies have confirmed that SCFA, including acetate, propionate, and butyrate, can maintain and promote the integrity of the intestinal structure by regulating the expression of barrier genes such as *ZO-1*, *OCLN*, and *CLDN* (Feng et al., 2018; Huang et al., 2021; Yue et al., 2022). Additionally, SCFA in the intestine can promote the expression of antibacterial peptide genes such as *REGIII*, *DEFB1*, and *DEFB4* through G-protein-coupled receptor 43, thus maintaining intestinal homeostasis (Zhao et al., 2018). Intestinal antibacterial peptides are mainly produced by Paneth cells in the small intestinal epithelium. They are important components of the innate immunity in the intestine and can maintain the homeostasis of the intestinal mucosa (Okumura and Takeda, 2018; Suzuki et al., 2022). Among them, REGIIIγ and defensins are common forms of intestinal antibacterial peptides and are already expressed in the intestines of calves at 3 d old (Zhu et al., 2020; Liu et al., 2023). Based on the aforementioned findings, we hypothesized that BCFA-DFL may regulate the integrity of the intestinal mucosa and the secretion of antimicrobial peptides by promoting the secretion of SCFA in the intestine, thereby maintaining intestinal homeostasis. Moreover, a few studies conducted in mice have revealed that BCFA can alleviate damage to the immune barrier caused by intestinal inflammation in mice. For example, Zhang et al. (2024) reported that supplementation with BCFA extracted from goat milk prevented the dextran sulfate sodium–induced increase in colonic *TNF-α* and *IL-1β* expression and the decrease in *IL-10* expression in a murine colitis model. Similarly, Ran-Ressler et al. (2011) demonstrated that feeding preterm rats BCFA with a composition and proportion matching that of vernix caseosa alleviated necrotizing enterocolitis by upregulating the expression level of *IL-10* in the ileum. These results were consistent with the regulatory effects of BCFA-DFL on the intestinal inflammatory genes of calves in our experiment.

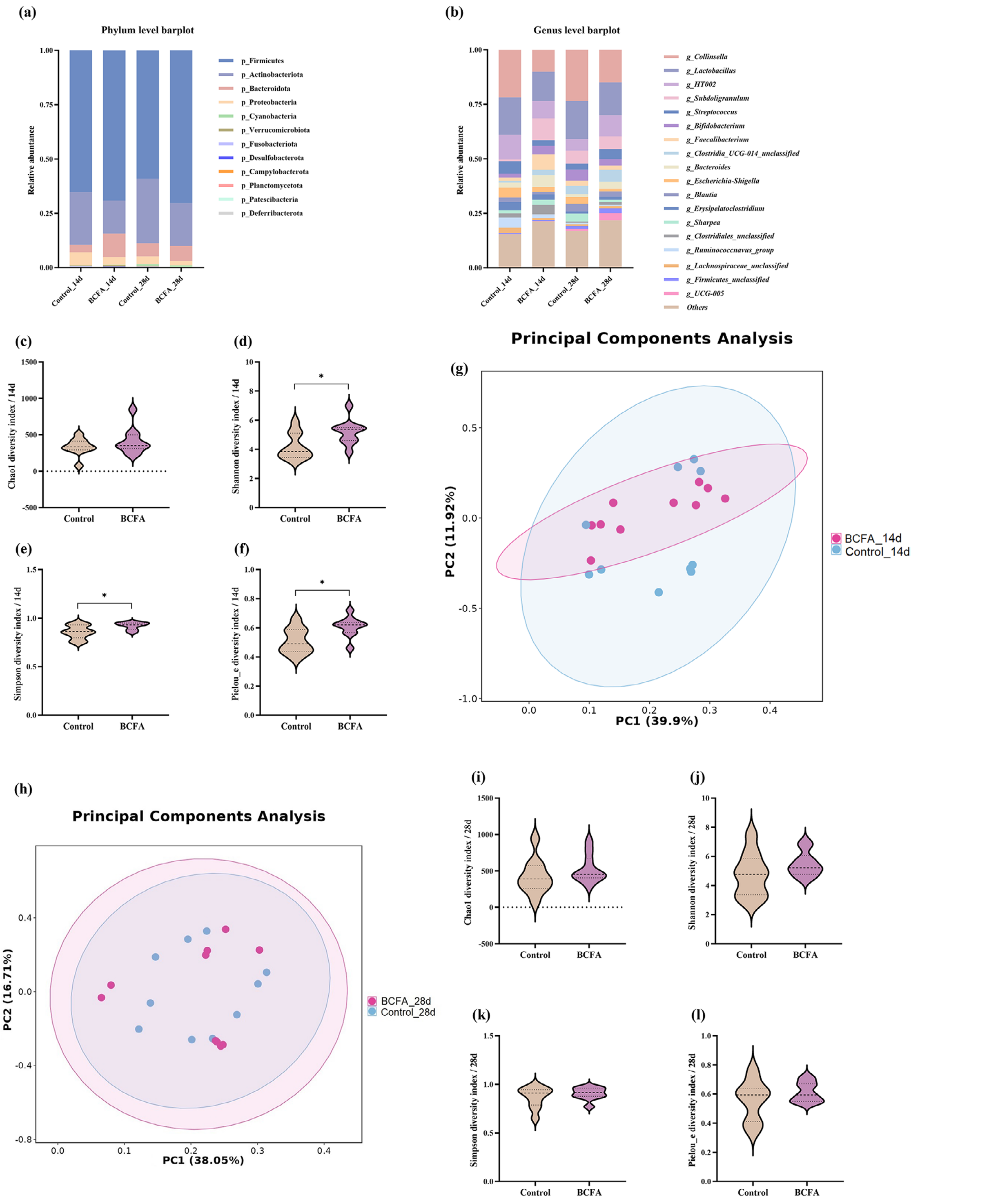

Figure 4. Effect of BCFA-DFL supplementation in whole milk on fecal microbiota in 14- and 28-d-old calves. (a) Distribution of fecal microorganisms in calves at the phylum level; (b) distribution of fecal microorganisms in calves at the genus level; (c–f) analysis of α diversity of fecal microbiota in 14-d-old calves by Chao1, Shannon, Simpson, and Pielou_e analysis, respectively; (g) principal component analysis (PCA) plots of β diversity of fecal microbiota (based on log-transformed relative abundance data using Euclidean distance as the underlying metric) in 14-d-old calves; (h) PCA plots of β diversity of fecal microbiota in 28-d-old calves. (i–l) analysis of α diversity of fecal microbiota in 28-d-age calves by Chao1, Shannon, Simpson, and Pielou_e analysis, respectively. In violin plots, the thin black dotted lines at both ends represent the quartiles, and the thick black dotted line in the middle represents the median. *Indicates a significant difference at $P < 0.05$ as determined by Wilcoxon test. PC = principal component.

Table 10. Effect of BCFA-DFL supplementation on the bacterial genus (relative abundances >0.1%) in the feces of dairy calves

Genus	Treatment[1] Control	BCFA	SEM	*P*-value
14 d				
Collinsella	21.81	10.02	3.180	0.17
Lactobacillus	17.21	13.44	3.180	0.63
HT002	11.31	7.93	2.242	0.97
Subdoligranulum	0.70	10.03	2.162	<0.01
Faecalibacterium	1.45	7.07	1.419	0.02
Clostridia_UCG-014_unclassified	0.84	2.44	0.429	0.01
Bacteroides	2.31	5.33	1.296	0.01
Sharpea	1.40	2.36	1.148	0.04
Peptostreptococcus	2.90	—		
Pseudoflavonifractor	0.08	0.33	0.059	0.02
Butyricicoccus	0.08	0.93	0.227	0.01
Negativibacillus	0.16	0.44	0.155	0.03
Prevotella_9	—	1.17		
Fournierella	0.04	0.47	0.115	0.01
Fusobacterium	0.64	—		
Paludicola	—	0.21		
28 d				
Collinsella	23.49	14.89	4.506	0.85
Lactobacillus	17.52	15.09	3.792	0.74
HT002	5.23	9.72	1.898	0.35
Subdoligranulum	5.92	5.79	1.325	0.68
Oscillospiraceae_UCG-005	0.97	3.30	1.060	0.04
Ruminococcus	0.46	0.93	0.179	0.03
Eubacterium_coprostanoligenes_group_unclassified	0.44	1.15	0.310	0.04
Christensenellaceae_R-7_group	0.37	1.21	0.376	0.03
Pseudoflavonifractor	0.35	0.82	0.279	0.03
Megasphaera	0.17	—		
Terrisporobacter	—	0.16		

[1]Control = calves fed whole milk containing 0.4 g of emulsifier (n = 12; 6 males, 6 females); BCFA = calves fed whole milk containing 4 g of BCFA-DFL (n = 12; 6 males, 6 females); all the treatments were carried out during the morning feeding.

Fecal Fermentation Parameters and Bacterial Communities of Calves

Calves from 0 to 1 mo old are in the period of gut microbiota colonization (Dias et al., 2018; Jessop et al., 2024). The rumen is not fully developed, and calves mainly rely on the intestine for the metabolism and absorption of nutrients (Kumar et al., 2021). During this stage, the gut microbiota is primarily involved in the intestinal metabolic processes of AA, carbohydrates, and energy, while also producing SCFA that serve as an energy source for intestinal epithelial cells and contribute to the maintenance of intestinal barrier integrity, as well as the homeostasis of inflammatory and immune responses in the internal milieu (Song et al., 2018; Mirzaei et al., 2021; Zhang et al., 2022). This period is also a high-incidence stage for diseases such as calf diarrhea, which is often accompanied by a disorder of the gut microbiota structure (Chen et al., 2022; Huang et al., 2024). Therefore, the colonization of gut microbiota in the early stage of life is of great significance for the growth, development, and health status of calves (Malmuthuge et al., 2015; Gomez et al., 2017). At 14 d of age, the proportion of butyric acid in fecal SCFA was significantly elevated in calves belonging to the BCFA group. As the most critical component of intestinal SCFA, the butyric acid content in calf feces exhibits a strong negative correlation with the severity of diarrhea. Notably, calves with lower fecal scores tend to harbor a more abundant population of butyric acid-producing bacteria in their feces (Song et al., 2018; Kumar et al., 2021). Most of the genera that caused differences in the microbiota structure between the 2 treatment groups in the fecal samples have the function of producing butyric acid, including *Subdoligranulum*, *Faecalibacterium*, *Bacteroides*, *Butyricicoccus*, *Sharpe*, *Clostridia UCG-014 unclassified*, and *Pseudoflavonifractor*, and their relative abundances were all higher in the BCFA group. Among them, the relative abundances of *Subdoligranulum* and *Faecalibacterium* had the greatest differences, and they have been reported to be able to prevent intestinal inflammation and are the marker genera reflecting the intestinal health status of calves (He et al., 2022; Nan et al., 2024). Furthermore, Xin et al. (2021) demonstrated that the fecal relative abundance of *Subdoligranulum* was significantly higher in healthy calves than in those with diarrhea, and this genus exhibited a positive correlation with the fecal concentrations of iso-C14:0, iso-C15:0, and anteiso-C15:0. As a class of lipid components of microbial cell membranes, BCFA have been verified to facilitate the intestinal colonization of commensal microbes (Ran-Ressler et al., 2011). Future investigations could use in vitro culture systems combined with targeted metabolomics approaches to elucidate whether BCFA directly enhance the colonization of *Subdoligranulum* and other differentially abundant microbial taxa. In addition, other genera with higher relative abundances in the feces of the BCFA group, such as *Bacteroides*, *Butyricicoccus*, *Sharpe*, *Clostridia UCG-014 unclassified*, and *Pseudoflavonifractor*, can directly or indirectly promote the production of SCFA mainly composed of butyric acid in the intestine and maintain intestinal homeostasis (Louis and Flint, 2009; Kamke et al., 2016; Chen et al., 2023; Wang et al., 2019; Trachsel et al., 2018). Moreover, *Peptostreptococcus* and *Fusobacterium* are often positively correlated with diarrhea in the early stage of calves (Hennessy et al., 2021; Chen et al., 2022; Claus-Walker et al., 2024), and they were only present in the samples of the control group. In contrast, *Prevotella_9*, as a potential probiotic for preventing calf diarrhea (Chen et al., 2022), was only found in the samples of the BCFA group. These results further reflect the differences in the microbiota structure caused by the differences in diarrhea incidence. Furthermore, we also observed that the calves in the BCFA group had higher α diversity indexes, namely Shannon, Simpson, and Pielou_e, in their feces at 14 d of age. Oikonomu et al. (2013) found that the increase in the fecal microbial diversity in the early stage of calf life was positively correlated with the increase in BW, and the results of the growth performance we measured also reflected the same trend. In addition, the fecal microbial diversity of sick (gastrointestinal or respiratory) calves is lower than that of healthy calves (Li et al., 2023; Jessop et al., 2024), which is also consistent with the lower health status of calves in the control group. At 28 d of age, fecal samples from calves in the BCFA treatment group exhibited elevated concentrations of total SCFA, along with higher proportions of isobutyrate and isovalerate. Concomitantly, calves in BCFA group also displayed enhanced apparent nutrient digestibility, and previous studies have also found that the increase in the apparent digestibility of nutrients contributes to an increase in the level of SCFA in feces (Zhao et al., 2020). In addition, fecal isobutyrate and isovaleric acid have also been identified as differential metabolites that reflect the health and diarrheal status of calves (He et al., 2022). Throughout the entire experimental period, calves in the BCFA treatment group exhibited lower diarrhea incidence, shorter diarrhea duration, and less frequent medication administration for diarrhea than those in the control group. The genera that showed differences in the fecal microbiota of the 2 treatment groups were mostly beneficial bacteria in the intestine, and their relative abundances were higher in the feces of calves in the BCFA group. Among them, *Pseudoflavonifractor*, *Oscillospiraceae_UCG-005*, *Christensenellaceae R-7*, and *Ruminococcus* have functions similar to those of probiotics, which can produce SCFA and prevent intestinal inflammation (Li et al., 2021; Gao et al., 2022; Chuang et al., 2022; Jin et al., 2023). *Eubacterium_coprostanoligenes_group_unclassified* has been reported to strengthen the integrity of the intestinal mucosal barrier and reduce intestinal susceptibility (Bai et al., 2024). Finally, *Megasphaera* and *Terrisporobacter* were only present in the samples of the control group and the BCFA group, respectively. *Megasphaera* can prevent acute diarrhea and promote intestinal development (Carey et al., 2021), and the relative abundance of *Terrisporobacter* is related to a lower inflammatory state of the body (Zhang et al., 2020; He et al., 2023). The colonization of the gut microbiota in calves has always been dynamic (Du et al., 2023). The individual genera with lower relative abundances in the 2 groups can reflect the short-term differences in the colonization of some genera, but overall, the gut microbiota of calves in the BCFA group was in a healthier state at 28 d of age. Moreover, multiple studies have proposed that in the early stage of calf birth, host (calf) immunity is the core driving factor for the colonization of the microbiota, and immune homeostasis plays a protective role in the colonization of intestinal microorganisms (Gamsjäger et al., 2023; Cai et al., 2024; Nishihara et al., 2024). Therefore, combined with the findings that BCFA-DFL positively regulates the expression of intestinal immune-related genes in calves in the present study, we conclude that the early colonization of beneficial gut microbes is favorably modulated by host immune regulation. Fecal isobutyric acid and isovaleric acid are primarily generated via the microbial fermentation of branched-chain AA, including leucine and valine, and are biomarkers of protein fermentation in the gut (Taormina et al., 2020; Wang et al., 2025). Despite their underrepresentation in investigations focusing on intestinal SCFA in calves, these 2 fatty acids are comparable to other SCFA in their capacity to maintain intestinal homeostasis (Lee et al., 2020). Furthermore, isobutyric acid can be oxidized to butyric acid to supply energy for colonic epithelial cells when intestinal butyrate availability is insufficient (Blachier et al., 2007). Accumulating evidence has indicated that supplementing isobutyric acid or isovaleric acid to calves before and after weaning can improve their growth performance (Liu et al., 2016; Wang et al., 2017). In addition to the reasons related to the health status, many studies have shown that the increase in the levels of isobutyric acid and isovaleric acid in feces also results from the increase in the total protein content in the diet (Liu et al., 2020b; Chi et al., 2024). In the present experiment, given the absence of differences in DMI between the 2 experimental groups, we hypothesize that calves in the BCFA group had greater quantities of dietary proteins and peptides available for microbial metabolism in the intestine, particularly in the large intestine. Based on our results, we believe that BCFA-DFL can promote the early colonization of beneficial bacteria in the intestine, improve the intestinal health of calves by promoting the secretion of SCFA mainly composed of butyric acid, and reduce diarrhea frequency.

However, the present study is not without limitations. All intestinal tissue samples used to investigate intestinal development and immune function were exclusively obtained from male calves. This experimental design was based on practical considerations in ruminant production practices. Consequently, the effects of BCFA on intestinal development in female calves cannot be elucidated from the current dataset. To validate the generalizability of the findings reported herein, future research should include female calves as experimental subjects. It should be noted that the fecal samples used for microbiome analysis in each batch were collected at a single time point. Although this allowed for comparison of treatment effects at that specific age, it may overlook potential differences in the microbial structure of calves at different time points within the same day. Therefore, we interpret the current results with caution, and future studies with increased sampling frequency are warranted to obtain more representative and dynamic profiles of the gut microbiota.

CONCLUSIONS

This study demonstrates that early-life supplementation with BCFA-DFL exerts a beneficial effect on growth metabolism and intestinal health status in calves. Specifically, BCFA-DFL supplementation can prevent and alleviate the occurrence of calf diarrhea, while positively regulating the expression of genes associated with intestinal mucosal integrity, inflammation, and immunity. Furthermore, this dietary intervention facilitates the early colonization of beneficial gut bacteria and promotes colonic fermentation in calves. These indicate that the growth-promoting effect of BCFA-DFL on calves is primarily attributable to its ability to enhance intestinal health status, a process that is largely mediated synergistically by intestinal immune factors and the gut microbiota.

NOTES

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Nonstandard abbreviations used: ALB = albumin; ASV = amplicon sequence variant; BCFA = branched-chain fatty acids; BCFA-DFL = BCFA derived from lanolin; EE = ether extract; H&E = hematoxylin and eosin; HDLC = high-density lipoprotein cholesterol; HP = haptoglobin; LDLC = low-density lipoprotein cholesterol; LFA = lanolin fatty acids; NEFA = non-esterified fatty acids; OR = odds ratio; PC = principal component; PCA = principal component analysis; SAA = serum amyloid A; SCFA = short-chain fatty acids; TG = triglyceride; Trt = treatment.

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